

Eye training games for children with low vision: A design-based research study

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ABSTRACT: This study investigated the design processes for developing eye training materials for children with low vision (CLV) using computer game applications based on eye movement tracking to enhance their vision skills. The primary aim was to create interactive eye training materials tailored to improve CLV's vision abilities. Employing a design-based research methodology, the study followed four distinct phases: analysis, development, evaluation and testing, and documentation and reflection. Initially, a game-based eye training program was developed. This program was then tested with CLV to gather relevant data for refining the design of eye-gaze games. Data were meticulously collected from a diverse group of stakeholders, including parents, special education teachers, ophthalmologists, and expert faculty members, using various methods such as interviews, focus group meetings, usability testing, and direct observations. The results provided critical insights into the design processes required to create a practical and usable game-based eye training program for CLV. These findings highlighted that children with low vision could benefit significantly from eye-tracking technology, using it to enhance their scanning, focusing, and tracking skills. Overall, the study suggests that eye-tracking technology offers a promising tool for developing vision-improving interventions for children with low vision, thus contributing valuable knowledge to the field of vision rehabilitation.

Keywords: Eye training games, Vision rehabilitation, Low vision, Eye-tracking technology, Design-based research

1. Introduction

According to Keeffe (1995), there are two categories of vision; one is normal vision and the other is visual impairment. People with normal vision are able to do all visual tasks well, whether from a near or distant distance. On the other hand, people with visual impairments have limited vision due to any condition, such as an accident or medical history. This group can experience total blindness, which is the complete inability to see light, or low vision, characterized by limited vision. World Health Organization (2007, 2012) describes low vision as visual acuity in the better eye ranging from 5 percent to less than 33.33 percent.

According to data from the World Health Organization (2012), 285 million people worldwide suffered from visual impairments in 2010; over 246 million of these individuals had low vision. Additionally, around 19 million of 285 million children aged 0-14 years had visual impairments, while approximately 18 million were those with low vision. Because better vision skills invariably enable children to perform their daily activities with less effort and enhance their quality of life, early interventions are necessary to maintain and improve children's functional vision skills (Gothwal et al., 2003; Keeffe, 1995; Strong et al., 2008). In addition, early visual rehabilitation to improve functional vision abilities can give them an opportunity to participate in school activities alongside their peers who have normal vision (Ganesh et al., 2013). Given that visual impairments can now be addressed with the proper care and support, eye training is necessary for these children to have fulfilling lives (Singh & Kumar, 2013; Strong et al., 2008).

Computer-based technologies are among the leading solutions in overcoming children's visual problems or helping children with low vision (CLV) enhance their vision (Achtman et al., 2008; Donmez & Cagiltay, 2019). It appears that game-based eye training systems can incorporate eye-tracking technologies (Donmez, 2023; Fikar et al., 2018). Previously, Kasprowski et al. (2016) employed a vision training system based on eye-tracking technology for CLV and found that eye-tracking-based games contribute much to such children's vision therapies. Eye-tracking technology, as an emerging technology, can also be employed in visual field assessment for CLV (Simkin et al., 2019). Also, it may be used as an on-screen visual search activity by tracking children's eye movements (Ivanov et al., 2018). Ultimately, eye-tracking technology is expected to obtain reliable and objective data on the eye movements of CLV (Kooiker et al., 2014). Besides, Kooiker et al. (2016) came to the conclusion that children with visual impairments might have their visual profiles analyzed using eye-tracking equipment. As a result, it may be practical to see how eye-tracking technology might be used to create eye

training and evaluation environments for CLV using stable, unbiased, and objective data regarding their eye movements.

In order to enhance their visual abilities, CLV must engage in eye training exercises (Vervloed et al., 2006). Maintaining everyday activities and enhancing the quality of life are also positively impacted by strengthening eyesight abilities, as noted by Keeffe (1995), Pasian et al. (2012), and Strong et al. (2008). Additionally, academic skills, including reading, writing, spelling, and math proficiency, can be enhanced via visual training exercises (Helveston, 2005). Since reading is one of the essential issues for CLV, academic progress is known to be strongly impacted by reading skills in particular (Lan et al., 2009; Vervloed et al., 2006). In other words, raising a child's reading proficiency can raise their academic standing. It requires controlling eye movements, along with tracking, focusing, and scanning skills (Carney et al., 2003). Besides, CLV do better in school when eye training is combined with early visual rehabilitation (Ganesh et al., 2013).

Teaching children with special needs can be made easier and more comfortable by utilizing technology as an intervention, such as technology-enhanced computer games and mobile apps (Chen et al., 2019; Thoo et al., 2023). Additionally, gaming consoles like the Nintendo Wii and Xbox can use body movements for controlling games in educational settings to provide benefits for children with motor impairments (Cagiltay et al., 2019). Besides, it is obvious that the influence of computer games on children's development is significant, and children with disabilities can also benefit from them (Chen et al., 2019).

With features like color, contrast, and light sensitivity, computers equipped with eye-tracking technology can be utilized to support eye training programs to improve the eyesight of people with low vision. This improvement can be ensured by enhancing some known vision skills like scanning the environment, focusing on the objects, tracking the moving objects, and improving the control over eye movements. The enhancements to these skills are also likely to result in both higher academic achievement and a higher quality of life for people with low vision. However, the current research and practice seem to miss the potential applications of eye-tracking technology in eye training programs. This technology can provide observable improvements in the vision capability of people having low vision compared to traditional practices for eye training which are based on subjective data and depend on the practitioner. In the end, the study aimed to create a program for eye training, consisting of eye gaze gaming applications for CLV, and to investigate the potential of eye-tracking technology, a revolutionary technology, in CLV rehabilitation.

2. Methodology

While attempting to steer the ultimate purpose of the study, we employed a design-based research methodology. The selection of design-based research as the methodology for this study was motivated by several key advantages it offers over other research methods. According to Wang and Hannafin (2005), design-based research not only facilitates the steering of the study's ultimate purpose but also provides a unique opportunity to integrate design, research, and practice. It was structured to utilize instructional development following design, development, and evaluation steps (Richey et al., 2004). By embracing design-based research, this study is better equipped to ensure that the theoretical aspects are effectively translated into practical, real-world solutions, making it a fitting choice for comprehensively examining the research objectives and implementing findings.

2.1. Participants

This study followed a design-based research methodology in four phases. Each phase of this study involved different groups of participants. While selecting participants, convenience and purposeful sampling was used to ensure that selected participants had knowledge about the topic (Creswell & Creswell, 2018).

At the beginning of this study, we obtained the necessary approvals from the Human Subjects Ethics Committee and the Applied Ethics Research Center at Middle East Technical University. In addition, we obtained permission from the Ministry of National Education of Turkey to collect data from children. All of the participants were informed about the study's objectives, and they were asked to sign consent forms to confirm their voluntary participation. Parents also signed consent on behalf of their children and were with them during the explanations and data collection. After the data was collected, participants were given debriefing forms with contact information for further questions.

There were various participants who were included in the analysis phase. The first group of participants consisted of four faculty members who were experts in the field of visual impairments. The second group included four teachers who worked in a school for students with visual impairments. The last participant was an ophthalmologist who has experience with people with low vision. All of these participants joined in the needs analysis, and tested the initial prototype of the eye training program.

The development phase of this study consisted of three iterative development processes. They were the initial prototype, second version, and final product. We worked with a faculty member who is an expert in the education of visually impaired children while developing storyboards for the eye training program. Besides, feedback for this eye training program was collected from two other faculty members, one from instructional technology and the other from special education. Finally, a formative evaluation was followed with four faculty members and four teachers specializing in visual impairments to apply necessary improvements to the program. Therefore, the second version of the program for eye training was created. Following the creation of the second version, focus group meeting and pilot test were applied as formative evaluations. Three faculty members from the field of education for visually impaired children participated in the focus group meeting. Then, a pilot study of eight sessions per day was conducted with an eight-year-old child with low vision. The purpose of gathering all of these data was to finalize the eye training program.

Table 1. Demographics of the children who participated in this study

Nu	Gender	Age	At-home eye exercise (Y/N)	Their diagnosis	Visual acuity of right eye (%)	Visual acuity of left eye (%)
P01	F	12	Y	Hereditary macular dystrophy	16	20
P02	M	10	N	Cone-rod dystrophy	60	60
P03	M	7	Y	Albinism	10	16
P04	M	18	N	Congenital retinoschisis	5	5
P05	M	11	Y	Cone dystrophy	10	16
P06	F	14	N	Congenital retinoschisis	5	4
P07	M	9	Y	Cortical visual impairment	0.1	2
P08	M	2	Y	Ocular Albinism	10	10
P09	F	13	N	Stargardt macular dystrophy	5	5
P10	M	9	N	Congenital cataract	16	10
P11	F	8	Y	Glaucoma	5	5
P12	M	12	N	Stargardt macular dystrophy	5	5
P13	M	10	N	Leber congenital amaurosis	5	5
P14	M	12	N	Congenital cataract retinoschisis	10	10
P15	F	13	Y	Albinism	10	10
P16	F	14	N	Macular dystrophy	5	5
P17	M	18	N	Degenerative Myopia	5	5
P18	M	8	N	Leber congenital amaurosis	1	1
P19	M	10	Y	Hereditary macular dystrophy	16	16
P20	M	6	N	Disc Hypoplasia	1	20
P21	F	6	Y	Hereditary macular dystrophy	5	5
P22	F	17	N	Retinitis pigmentosa	5	5
P23	F	15	N	Retinitis pigmentosa	10	5
P24	M	6	N	Retinitis pigmentosa	5	5
P25	F	10	N	Retinoschisis	30	30
P26	F	16	N	Amblyopia	16	40
P27	M	11	Y	Retinopathy of prematurity	5	5
P28	F	18	N	Albinism	20	20
P29	F	12	Y	Cerebral Visual Impairment	60	60
P30	M	10	Y	Ocular Albinism	16	16
P31	M	11	Y	Hereditary macular dystrophy	10	10
P32	F	11	N	Amblyopia	30	30
P33	F	5	N	Hereditary macular dystrophy	10	10
P34	M	17	N	Cerebral Visual Impairment	60	20
P35	F	8	N	Optic atrophy - Didmoad syndrome	60	50
P36	F	14	N	Optic atrophy - Didmoad syndrome	20	30

During the evaluation and testing phase, usability testing was carried out by recording participant usage and doing observations. We chose the participants by using purposeful and convenient sampling. Participants in this study were children who visited the low vision unit of a university for a duration of eight weeks. This unit was

connected to the ophthalmology department, and responsible for conducting research on low vision and providing rehabilitation services. Thirty-six of these children agreed to be involved in this study. Each child used the eye training system that was explicitly developed in the scope of the current study. It took twenty minutes for each child to experience the system, and the demographics of these children were presented in Table 1.

The age range of the children in this study spanned from two to eighteen years old, with nearly one-third falling between the ages of 10 and 12. On average, the participants were around eleven years old. Among the participants, 19 were male and 17 were female. Their eye problems encompassed a wide range of diagnoses and visual acuity levels, with hereditary macular dystrophy, albinism, and retinitis pigmentosa being the most common diagnoses. Visual acuity values, expressed as percentages, indicated the clarity of vision in each eye, with higher percentages representing stronger visual acuity and lower percentages indicating poorer visual acuity. The data underscored the significant challenges these children face due to a spectrum of visual impairments, impacting their daily lives and necessitating specialized care and support.

Games created to improve scanning and tracking skills could help tackle the visual challenges these children face. Even though each diagnosis comes with its own set of problems, the specific skills targeted by these games could improve visual abilities across different conditions. Engaging in these activities allows children to hone their scanning and tracking skills, potentially boosting their overall visual abilities and improving their quality of life. This study also included interviews with the parents of the CLV to collect their suggestions and comments about the eye training system.

2.2. Design processes of the study

According to Reeves et al. (2004), design-based research involves four phases, namely analysis, design, evaluation and testing, and documentation and reflection, and they were followed in the current study (see Figure 1). Fraenkel et al. (2011) highlight that qualitative data can be obtained through various methods, such as focus groups, observations, and interviews. In this study, focus group meetings, interviews, expert insights, observations, and usability testing were used as our data collection tools. This also aligns with the view of Creswell and Creswell (2018), who emphasize the importance of using multiple data sources.

Figure 1. Design processes of this study

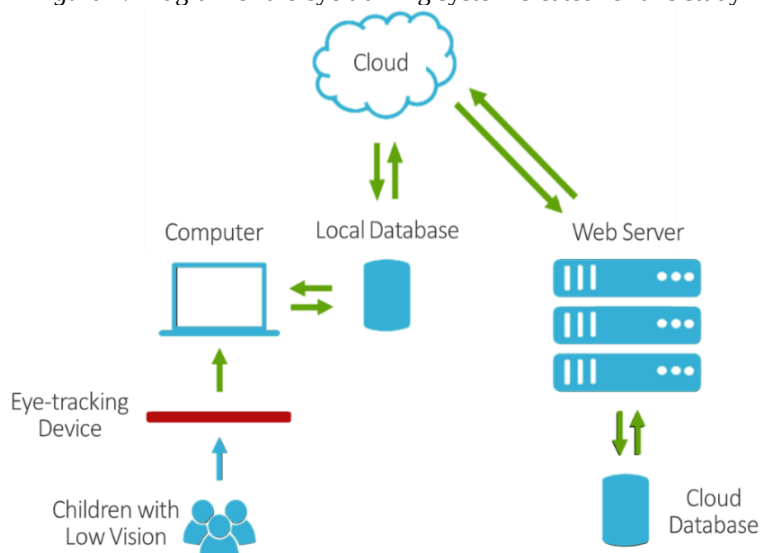
Phase 1	Phase 2			Phase 3	Phase 4
Analysis of Practical Problems by Researchers and Practitioners	Development of Solutions with a Theoretical Framework			Evaluation and Testing of Solutions in Practice	Documentation and Reflection
1.1. Needs Analysis	2.1. Determining the Vision Skills to Develop Training	2.3. Development of First Prototypes	2.6. Formative Evaluation of Prototypes	3.1. Summative Evaluation of Eye Training Program	4.1. Refining Design Processes
1.2 Pre-prototype Tests (Review of Theoretical and Practical Applicability)	2.2. Preparing the Storyboards	2.4. Development of Second Prototypes			
		2.5. Development of Final Products of the Eye Training Program			

The first phase was all about the analysis of the problem. Initially, we kicked off a series of focus group meetings with three faculty members specializing in visually impaired students' education to perform a needs analysis. The primary role of focus group meetings was to obtain tentative ideas from these experts to introduce a robust eye movement exercise program for CLV. Then, we interviewed four teachers working in a school for children with visual impairments to comprehend their existing technological practices and needs while teaching CLV. Next, four faculty members specialized in visually impaired students' education were invited to a series of meetings to discuss the educational needs of CLV. We also interviewed an ophthalmologist to understand the medical aspects of such children's conditions. Finally, we recruited CLV in a pre-prototype eye training program trial that followed the interviews above.

In the second phase, we evaluated the solutions aired by the experts, generated a system flow diagram, and integrated the solutions into the prototypes of the desired program (see Figure 2). Firstly, we sought an expert opinion from a faculty member to identify the skills that CLV need to improve. Secondly, we prepared

storyboards for the initial prototype. Then, we determined the framework of the training program based on the storyboard revisions and feedback from two faculty members, one expert in the education of visually impaired students and the other in instructional technology. Next, we launched the iterative and incremental development process, at the end of which the initial prototype emerged. During the formative evaluation of the development phase, we performed eight interviews, four with special education teachers and four with faculty members in the relevant field. Then, we began developing the second version, considering formative evaluation notes. We introduced this version following a focus group meeting with three faculty members and their revision recommendations. In addition, we conducted eight trials of a pilot test with a child who had low vision, which helped us refine the program and turn it into a complete program. Overall, these development phases facilitated the emergence of the design processes necessary for creating the eye training program.

Figure 2. Diagram of the eye training system created for this study



This study's third phase, which covered the program's summative evaluation for eye training, was intended to be an evaluation and testing phase for the program. During this phase, we conducted usability testing using actual usage logs of the participants and observations to ensure the development of a usable eye training program aimed at improving CLV's vision skills. A total of 36 CLV played four eye movement games, aiming at enhancing their scanning, focusing, and tracking skills, for a total of 20 minutes, spending 5 minutes on each game (see Figures 3 and 4).

Figure 3. The games for enhancing scanning-focusing skills

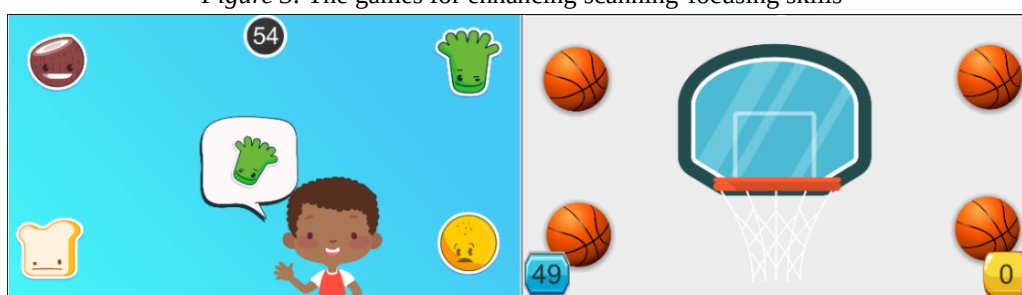


Figure 4. The games for enhancing scanning-tracking skills



In order to evaluate their eye movements in light of their medical history and demographic characteristics, we continuously recorded their eye movements, scores, and playing habits during the usability testing (see Figure 5). To find out what they thought of the training program, we also conducted interviews with their parents. We interviewed with the ophthalmologists in four more interviews in addition to the four we did with the faculty members to get their feedback on the eye training program.

Figure 5. Data collection setting of the study



The final phase of the study focused on documentation and reflection to generate relevant design processes. During this phase, we finalized and documented the design processes based on the findings from the evaluation and testing phase.

3. Results

The design processes that emerged during this study were generated during the four phases, namely analysis, development, evaluation and testing, and documentation and reflection.

3.1. Analysis phase

The needs analysis was conducted with four faculty members, four teachers, and an ophthalmologist. According to this analysis, it could be said that participants were both capable of and were willing to experience any practical, effective, and efficient computer-based technology. They knew that it was crucial for CLV to do eye exercises, and any technology could be used for this purpose. In this study, the Tobii Eye Tracker 4C was used to capture the eye movements of CLV. During the interviews with these participants, they were informed about the potential of using eye-tracking technology for CLV as an eye training program. By considering the information provided for them about eye-tracking technology, they pointed out that it is necessary to focus on the individual differences of CLV while developing the eye training program. They also underlined that doing eye exercises could help CLV enhance their reading and writing skills, which are also compulsory for maintaining their academic success in courses. Lastly, they explained that an eye training program developed for CLV should offer various settings to configure according to CLV's needs, color and contrast preferences/sensitivity, visual acuity, diagnosis, and visual field limitations.

Furthermore, pre-prototype tests were conducted to gather data about the needs and capabilities of CLV. Four visually impaired children from the school tried to use the eye training program's early prototype during the tests. Each child who used the early version of the eye training program was observed while using the program by sitting in front of the computer with an eye-tracking device. The observation notes indicated that the eye training program could collect CLV's eye movements. These children were able to focus on the objects displayed on the screen and track them. Besides, important considerations emerged in this phase. For instance, it was necessary to optimize the program's interface and functionalities to capture the eye movements of CLV, who could also have various disorders, and provide proper responses to them on the screen. Therefore, it became clear that the program's design should address the unique visual capabilities and challenges of this group. It also requires various features to facilitate object tracking and focusing by keeping engagement and efficacy. Incorporating these considerations into the program's design was crucial to ensure its accessibility and effectiveness for CLV.

3.2. Development phase

Prior to creating the CLV vision training program, a faculty member with expertise in low vision was interviewed to determine the vision skills that should be prioritized. It was chosen to concentrate on eye movement activities during the interview, particularly on scanning, focusing, and tracking abilities. After that, expert opinions were gathered through interviews with two faculty members. One faculty member specialized in low vision, while the other was an expert in instructional technology. Their insights led to the creation of storyboards for the development and design of the eye training program for CLV.

Next, using the storyboards as a guide, two games were created as the eye training program's initial prototype. The purpose of these games was to improve CLV's scanning, focusing, and tracking abilities. Four teachers and four faculty members participated in the formative evaluation following the creation of the first prototype. These interviews were done to learn about their opinions regarding the overall layout of the games, the issues that needed to be considered during revisions, and potential improvements to the eye training program. For example, when creating a vision training program for CLV, participants emphasized that using contrasting colors, providing options for color selection, and maintaining a simple design were crucial design considerations to take into account during development.

The eye training program's second version, consisting of four games, was then created using the formative evaluation results. In order to accomplish this goal, a pilot test involving a child with low vision was conducted by using observation and logging the actual usage of the games. Additionally, a focus group meeting with three faculty members who are experts in low vision was held in order to gather their feedback and suggestions regarding the games developed as part of the vision training program. The pilot test consisted of eight trials with this single child, each lasting 20 minutes, with 5 minutes dedicated to each of the four games. According to the findings, scores in the balloon popping (focusing) game varied from a minimum of 13 to a maximum of 26. Scores for the basketball (focusing) game ranged from 15 at the bottom to 54 at the top. The game of food collecting (tracking) has a maximum score of 25 and a minimum score of 11. Finally, in the basketball (tracking) game, the highest available score was 34, and the lowest available score was 12. The results demonstrated that the participant's scores displayed a comparatively incremental pattern among all games. Furthermore, the games' length and sequence were suitable for CLV. As a result, the pilot test's implementation environment and procedures were validated for CLV. Subsequently, the recommendations were used to create the final version of the eye training application, which included settings for altering the focus duration, object size, object color, and background color. Lastly, a vision training program based on eye-tracking technology was designed and developed for CLV using the data gathered from these procedures together with newly developed design processes.

3.3. Evaluation and testing phase

The summative evaluation included CLV, their parents, faculty members, and ophthalmologists to finalize the eye training program. There were 36 CLV during the vision training program's summative evaluation. CLV's usage logs and an observation checklist were used to collect the data. Each child played each game for five minutes. During this time, their performance was closely monitored, and their scores were recorded. 9 of 36 CLV could not complete all four games, while the rest could finish all four (see Table 2).

This table provides a comprehensive overview of several key aspects of the study. It details the visual acuities of participants' better eyes, showing how their vision levels correspond to their performance in the eye training program. The table also indicates whether participants completed all the games within the program. Additionally, it includes their scores in four distinct games: balloon popping (focusing), basketball (focusing), food collecting (tracking), and basketball (tracking). The table reveals a wide range of performance levels among participants, which is evident from their scores in the various games. While some participants achieved high scores across all games, others struggled to attain comparable scores, suggesting differing levels of proficiency in different visual tasks.

Furthermore, a potential correlation between visual acuity and game performance is evident. Participants with higher visual acuity percentages generally achieved higher scores in the games, confirming a positive relationship between visual capabilities and task performance. For example, none of the games could be finished by two children with low eyesight, whose better eye visual acuity was less than 5%. In contrast, the games were playable by 8 out of 12 children having 5% visual acuity. Additionally, 10 out of 11 children having visual acuity equal to or greater than 10%, but less than 20% were able to play the games. Besides, 9 out of 11 children having visual acuity equal to or greater than 20% were able to play the games. Thus, it was determined that children

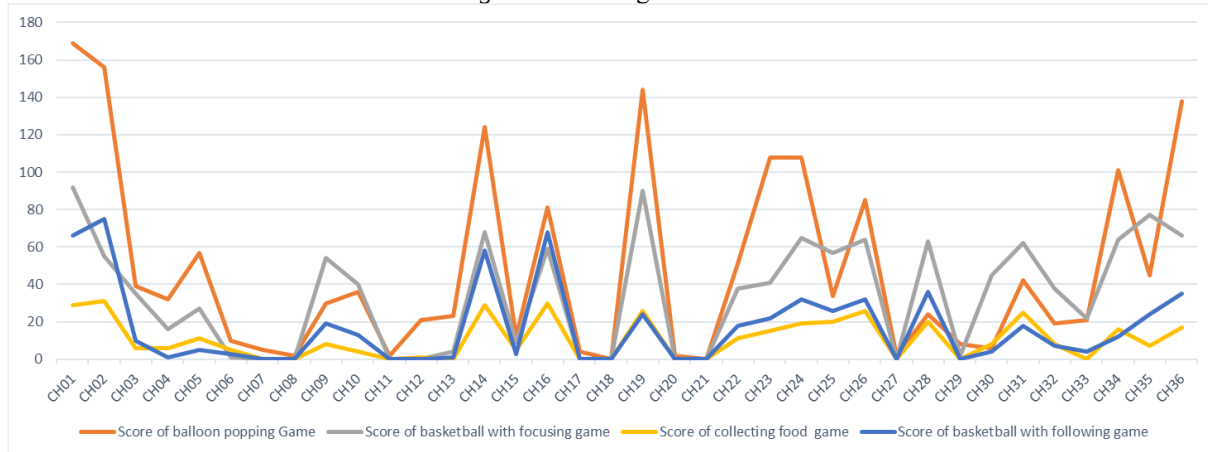
with low vision, defined as those having visual acuity of between 5% and 10%, should try the eye training program based on eye-tracking technology to see if it would work for them. On the other hand, children having visual acuity greater than 10% were likely to finish the eye training program utilizing eye-tracking technology. Moreover, these findings highlight the necessity of personalized approaches to visual training and intervention to address the unique needs of this population effectively. These findings are significant because they provide empirical support and quantification to what might be expected intuitively. At first glance, the findings may seem already obvious. However, as is well known, traditional rehabilitation methods for individuals with low vision do not employ technological approaches; instead, they typically use manual methods such as bead threading and tracking with hand signs. However, as evidenced by the findings of our pioneering study, children with various levels of visual acuity were able to utilize this innovative technological approach during the rehabilitation process. The results of the study provide concrete evidence, through experimental data, on the effectiveness of eye training program equipped with eye-tracking technology, specifically regarding the performance of different groups of children with varying levels of visual acuity.

Table 2. Scores obtained from CLV's usage logs

Number	Better eye's visual acuity	Completed All games (Y/N)	Balloon popping (focusing) Game score	Basketball (focusing) Game score	Food collecting (tracking) Game score	Basketball (tracking) Game score
P01	20%	Y	169	92	29	66
P02	60%	Y	156	55	31	75
P03	16%	Y	39	35	6	10
P04	5%	Y	32	16	6	1
P05	16%	Y	57	27	11	5
P06	5%	Y	10	1	5	3
P07	2%	N	5	-	-	-
P08	10%	N	2	-	-	-
P09	5%	Y	30	54	8	19
P10	16%	Y	36	40	4	13
P11	5%	N	2	-	-	-
P12	5%	Y	21	0	1	0
P13	5%	Y	23	4	0	1
P14	10%	Y	124	68	29	58
P15	10%	Y	12	9	5	3
P16	5%	Y	81	59	30	68
P17	5%	N	4	-	-	-
P18	1%	N	0	-	-	-
P19	16%	Y	144	90	26	24
P20	20%	N	2	-	-	-
P21	5%	N	0	-	-	-
P22	5%	Y	51	38	11	18
P23	10%	Y	108	41	15	22
P24	5%	Y	108	65	19	32
P25	30%	Y	34	57	20	26
P26	40%	Y	85	64	26	32
P27	5%	N	2	-	-	-
P28	20%	Y	24	63	20	36
P29	60%	N	8	2	-	-
P30	16%	Y	6	45	8	4
P31	10%	Y	42	62	25	18
P32	30%	Y	19	38	8	7
P33	10%	Y	21	22	0	4
P34	60%	Y	101	64	16	12
P35	60%	Y	45	77	7	24
P36	30%	Y	138	66	17	35

It was clear from the children with low vision's actual usage records that their game scores were similar to one another (Figure 6). There was a comparable trend between the players' scores in the balloon popping game and the scores in the other games. Despite the fact that the games' varying degrees of difficulty, these results show that players' game achievement rates were consistent.

Figure 6. CLV's game scores



Following the collection of CLV data, their parents were asked to provide insight into their experiences. Thirteen out of the thirty-six parents who participated in the interviews said that their children were practicing eye exercises. Conversely, twenty-three of them said that their children did not practice any eye exercises. Six out of thirteen parents who reported doing eye exercises with their children stated that they used objects to have their children focus on and track. Additionally, they were open to providing their children with at-home eye training routines that benefited from technology. Furthermore, they discovered that the eye training program utilizing eye-tracking technology was beneficial for their children to do eye exercises to improve their visual abilities.

We interviewed four ophthalmologists and four faculty members to get their opinions on using eye-tracking technology for low-vision rehabilitation in CLV. They agreed that the eye training program was good in improving functional vision, especially for tracking, focusing, and scanning skills. Most of the faculty members and ophthalmologists also suggested adding more games with different scenarios to make the program better.

3.4. Documentation and reflection phase

In this phase, the design processes of the eye training program were examined. During this review, the data gathered from the analysis, development, evaluation, and testing phases were used to make adjustments. Valuable information was gathered from each phase of this study to create an eye training program that better meets the needs of CLV. Besides, the design processes of the eye training program were documented in parallel with the purpose of this study. As a result, the findings of this study were combined to create the design processes for developing an eye training program for CLV.

4. Discussion and conclusion

This study's primary objective was to create a usable and game-based eye training program for CLV in order to enhance their ability to track, focus, and scan. The development of this eye training program was carried out with the contributions and suggestions of various participants, namely CLV, their parents, ophthalmologists, special education teachers, and faculty members from the field of instructional technology and special education. These contributions and suggestions to the development of this eye training program were crucial and valuable while reaching the final version of the program. The expertise and feedback from these participants could be combined and reflected in the eye training program's design and development processes. Therefore, we could ensure the effectiveness and usefulness of the program developed in this study.

Eye training exercises play a crucial role in enhancing CLV's vision skills, as highlighted by previous research (Vervloed et al., 2006). Moreover, the enhancement of vision skills has been shown to positively impact daily activities and overall quality of life, as noted by Keeffe (1995), Pasian et al. (2012), and Strong et al. (2008). The current study suggests that the eye training program can be considered an alternative form of game-based eye exercise for CLV in various manners. The game-based eye training program, utilizing data from eye-tracking device, may provide objective feedback about the experience of CLV and offer them the opportunity to engage in self-training for visual improvement anywhere and anytime. Moreover, there is a growing inclination among experts and practitioners to embrace technology in assisting children with low vision. Studies have associated visual training exercises with enhancements in academic skills such as reading, writing, spelling, and

mathematical proficiency (Helveston, 2005). Notably, proficiency in reading is particularly crucial for academic achievement among CLV (Lan et al., 2009), underscoring the importance of interventions targeting reading skills. Such interventions require the development of scanning, focusing, tracking skills, and effective eye movement control (Carney et al., 2003). Therefore, early visual rehabilitation through eye training holds significant promise for enhancing CLV's academic success (Ganesh et al., 2013). Thus, we may conclude that eye-tracking applications represent promising technological advancements that hold the potential to significantly benefit the lives and educational achievements of children with low vision, facilitating the enhancement of their vision skills (Ali et al., 2022).

Special education educators and practitioners can utilize the results of this study to adapt technology and offer vision training to CLV. The results of this study suggested that, in addition to the activities and evaluations carried out by professionals, it might be beneficial to give these children objective eye training. In order to practice eye training with CLV, special education teachers, vision therapists and ophthalmologists can benefit from an in-depth understanding of the design processes of this study.

Overall results will inevitably contribute to the relevant literature and guide the development of an eye training program for CLV. However, further research is still needed on this topic to elucidate how to integrate and utilize eye-tracking technology with robust, reliable data to monitor the eye movements of CLV to enhance their visual skills. The results of the study are anticipated to be beneficial for low-vision rehabilitation by practitioners, special education teachers, and instructional designers.

In further studies, to make the eye training program more individually adaptable, more children with different kinds of visual impairments could participate in the games designed for the program. Additionally, increasing the number of games and diversifying their settings could provide CLV with additional possibilities.

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